

Lecture 5: Atmospheres I

Composition, Structure, & Energy Balance

Planetary Systems

A horizontal, slightly curved blue line with a soft glow, representing the Earth's atmosphere as seen from space. It is centered horizontally and spans most of the width of the slide.

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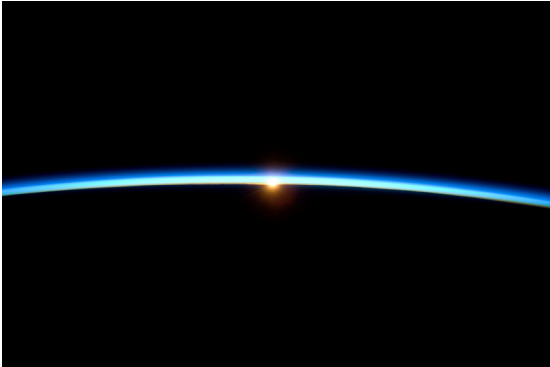
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Learning objectives

By the end of this lecture, you will be able to:

- Classify atmospheres as primary, secondary or tertiary
- Derive the scale height from hydrostatic equilibrium and apply it across the solar system
- Explain the greenhouse effect and the escape mechanisms that decide which atmospheres survive

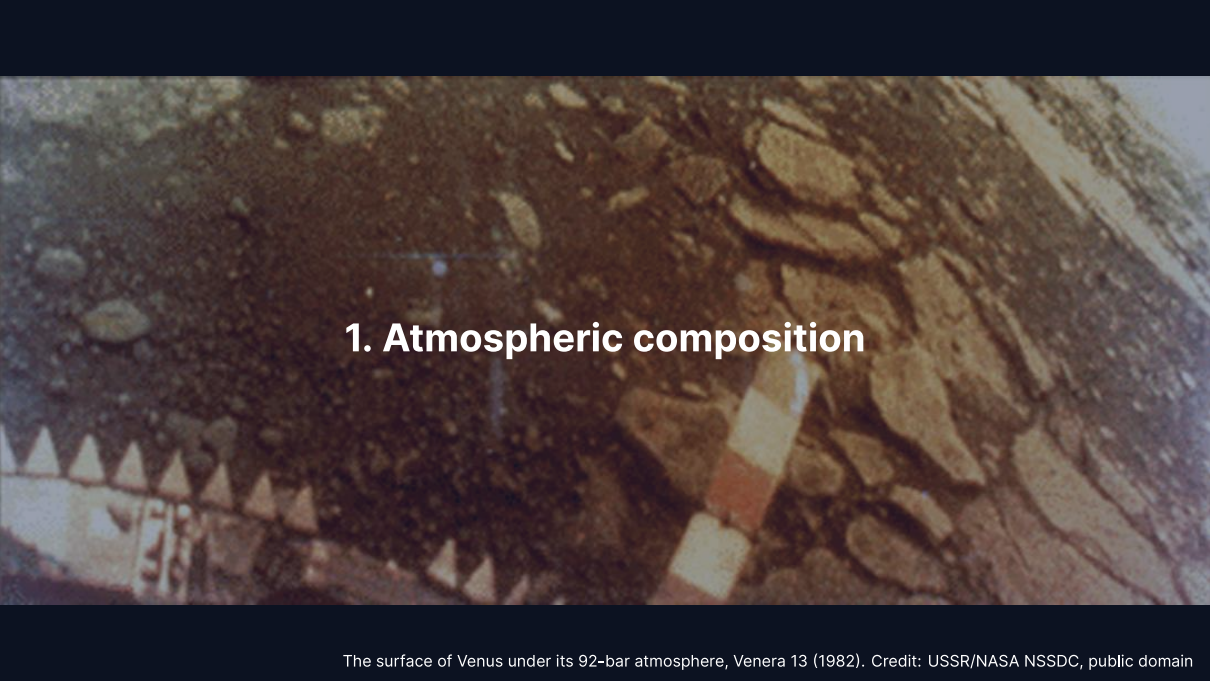
Recap: Lecture 4



Credit: NASA/ISS Expedition 21 crew, public domain

- Metal-silicate separation in magma oceans formed cores within ~30–60 Myr
- A dynamo needs a conducting fluid, convection and $R_m \gtrsim 10\text{--}100$
- Mars lost its dynamo between ~4.1 and 3.7 Ga, then its atmosphere

Today: The thin gaseous envelope that controls a planet's surface conditions.

A photograph of the Venus surface from the Venera 13 mission. The image shows a dark, rocky terrain with a color calibration target in the foreground. The target consists of a series of colored squares and triangles. The text "1. Atmospheric composition" is overlaid on the image.

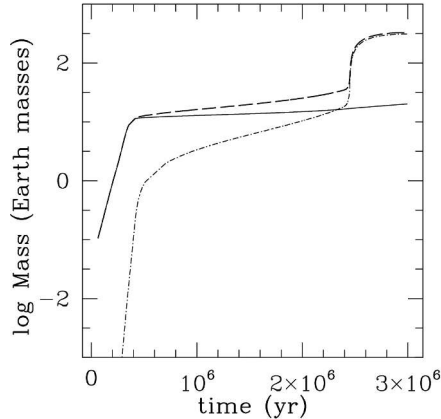
1. Atmospheric composition

The surface of Venus under its 92-bar atmosphere, Venera 13 (1982). Credit: USSR/NASA NSSDC, public domain

Primary atmospheres

Captured directly from the protoplanetary disk.

- Dominated by H_2 and He (solar composition)
- **Gas giants** keep massive H_2/He envelopes; **ice giants** only 10–20% of their mass
- Terrestrial planets are too small to retain primordial H_2/He



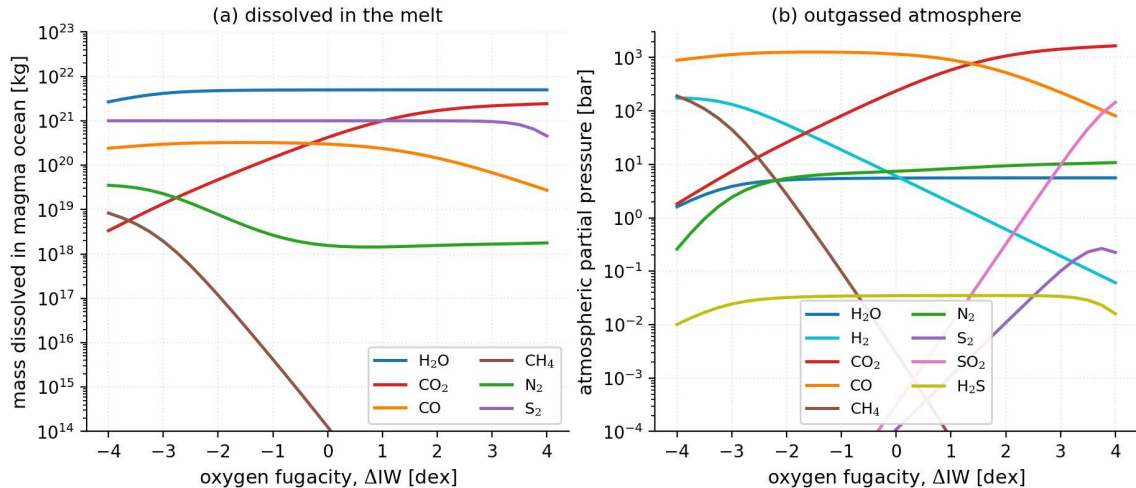
Growth of a giant planet by core accretion: runaway gas accretion builds the H_2/He envelope once it outweighs the core. Reproduced from Helled et al. (2014), Fig. 1; simulation from Lissauer et al. (2009).

A primary atmosphere needs a core of $\gtrsim 5\text{--}10 M_{\oplus}$ before the disk disperses in $\sim 3\text{--}10$ Myr.

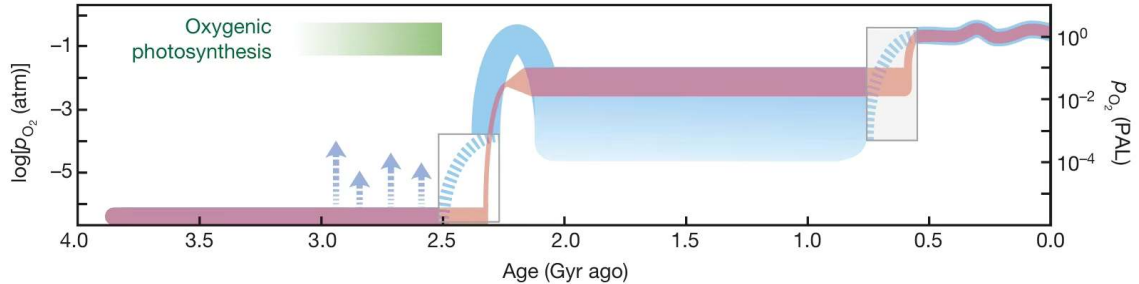
Secondary atmospheres

Produced by outgassing from the interior.

- Magma ocean degassing and volcanism release dissolved volatiles
- Speciation follows the mantle **oxygen fugacity**: oxidising gives CO_2 , H_2O , N_2 ; reducing gives H_2 , CO , N_2
- **Venus**: 96.5% CO_2 at 92 bar; **Mars**: 95% CO_2 at 6 mbar; **Titan**: N_2 (from NH_3) at 1.5 bar



Speciation of an outgassed atmosphere against melt oxygen fugacity: CO , H_2 and CH_4 when reducing, CO_2 and H_2O when oxidising. Course figure; model of Nicholls et al. (2024).



Earth's atmospheric oxygen through time: anoxic for ~2 Gyr, rising at the Great Oxidation Event near 2.4 Ga, modern levels in the Phanerozoic. Reproduced from Lyons et al. (2014).

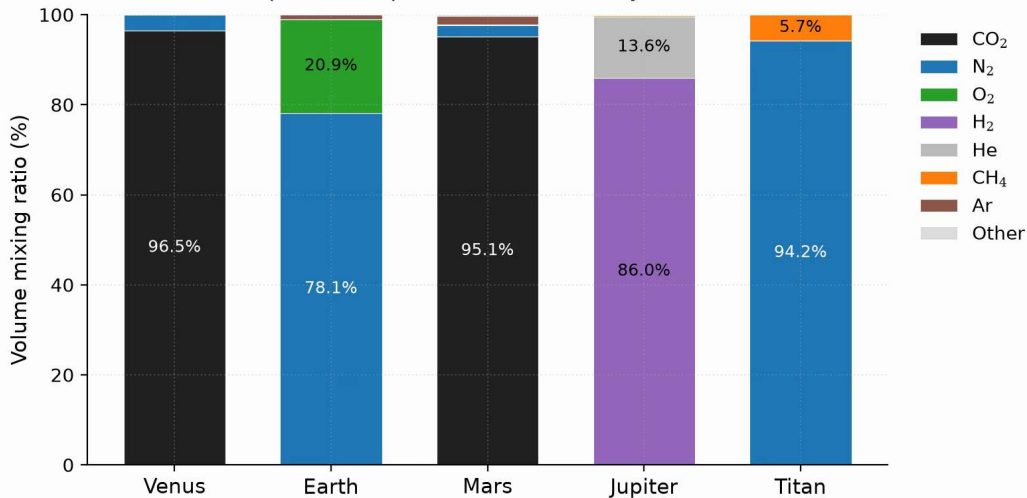
Earth's atmosphere is **tertiary**: the outgassed CO_2 and N_2 inventory was Venus-like, oxygenic photosynthesis added the 21% O_2 , and that O_2 is entirely biogenic and needs active life to persist.

Comparative atmospheric properties

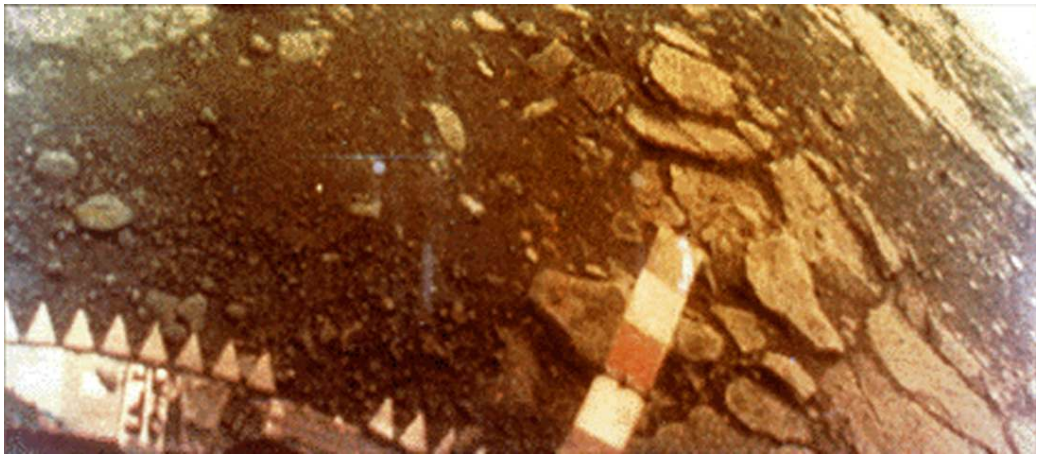
	Venus	Earth	Mars	Jupiter	Titan
P_{surf} (bar)	92	1.0	0.006	n/a	1.5
T_{surf} (K)	737	288	215	n/a	94
Dominant gas	CO ₂	N ₂	CO ₂	H ₂	N ₂
Secondary gas	N ₂	O ₂	N ₂	He	CH ₄
μ	43.4	28.97	43.3	2.2	28.6
Type	2*	3*	2*	1*	2*

Surface pressures span five orders of magnitude, temperatures run from 94 to 737 K.

Atmospheric composition of five solar-system bodies



Atmospheric composition by volume for five solar-system bodies: three origins yield five distinct outcomes. Course figure.



The surface of Venus seen by *Venera 13* on 1 March 1982. Credit: USSR/NASA NSSDC, public domain.

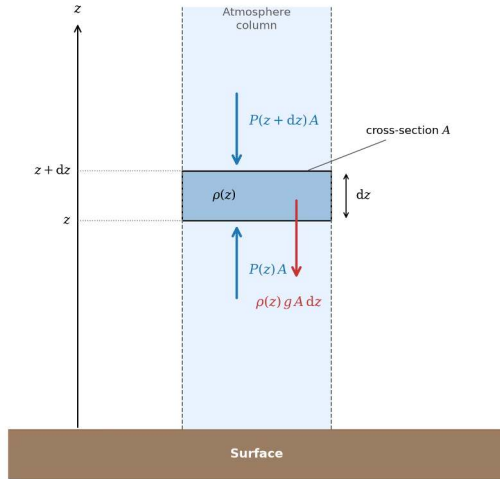
The lander survived 127 minutes at 92 bar and 737 K; the dense CO₂ atmosphere and cloud deck filter sunlight to a dim orange glow.



2. Hydrostatic equilibrium

The thin atmosphere of Mars above its limb, Viking 1 (1976). Credit: NASA/JPL

Hydrostatic equilibrium



Credit: Course figure

The net vertical force on a thin slab of area A and thickness dz vanishes:

$$P(z)A - P(z + dz)A = \rho g A dz$$

$$\frac{dP}{dz} = -\rho g$$

Each layer supports the weight of all gas above it.

The ideal gas law and barometric formula

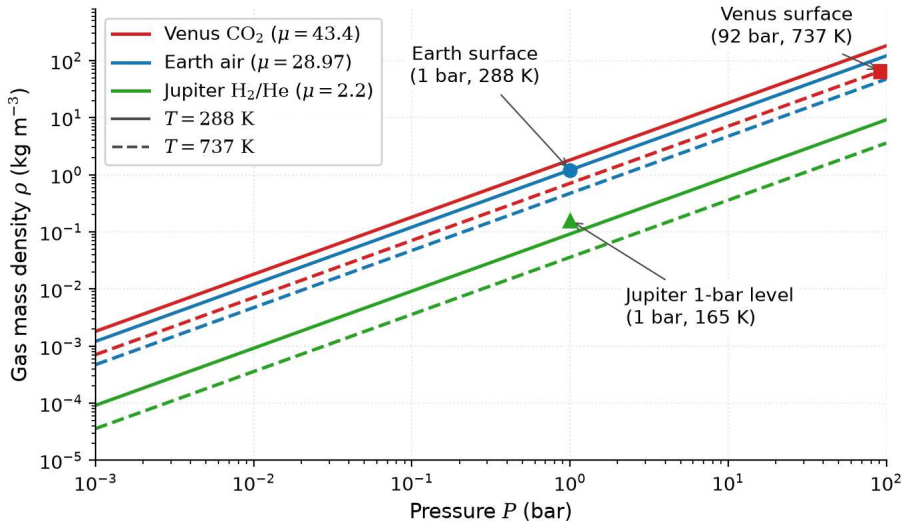
The ideal gas law $P = \rho k_B T / (\mu m_u)$ in $\frac{dP}{dz} = -\rho g$ gives

$$\frac{dP}{dz} = -\frac{\mu m_u g}{k_B T} P = -\frac{P}{H}$$

For an **isothermal** atmosphere (constant T):

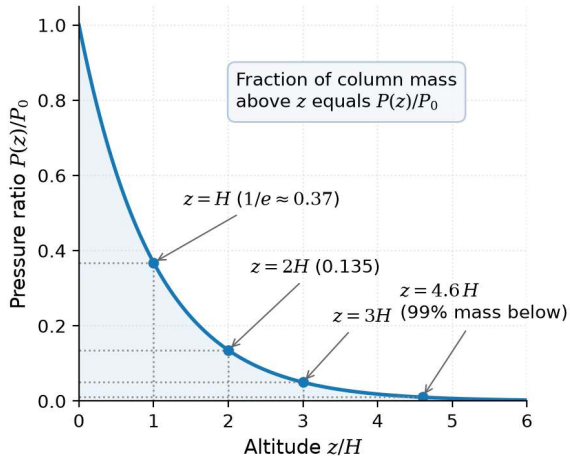
$$P(z) = P_0 \exp\left(-\frac{z}{H}\right)$$

with $H = k_B T / (\mu m_u g)$; the pressure drops by $e \approx 2.718$ each scale height.

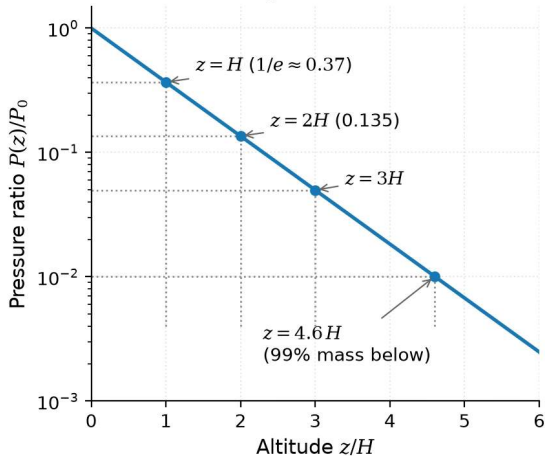


Ideal gas law in atmospheric form, $\rho = P\mu m_u / (k_B T)$: density scales with pressure at fixed T , heavier or colder gas is denser. Course figure.

(a) Linear scale



(b) Logarithmic scale



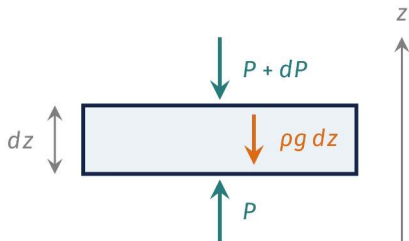
The barometric formula $P/P_0 = e^{-z/H}$: a factor e per scale height, 99% of the column mass below $4.6H$, a straight line on a logarithmic axis. Course figure.



3. Blackboard derivation: scale height

Goal and strategy

Goal: derive $H = k_B T / (\mu m_u g)$ from hydrostatic equilibrium and the ideal gas law.



- Balance the forces on a thin slab
- Replace ρ by P with the ideal gas law
- Read off the length scale, integrate, compute

One length, H , sets how fast pressure falls with height.

Deriving the scale height

Start from:

$$\frac{dP}{dz} = -\rho g \quad \text{and} \quad P = \frac{\rho k_B T}{\mu m_u}$$

Express ρ in terms of P :

$$\rho = \frac{P \mu m_u}{k_B T}$$

Substitute:

$$\frac{dP}{dz} = -\frac{\mu m_u g}{k_B T} P \equiv -\frac{P}{H}$$

The atmospheric scale height

$$H = \frac{k_B T}{\mu m_u g}$$

- **Higher T** → larger H : hotter gas extends further
- **Heavier molecules** (larger μ) → smaller H : harder to loft
- **Stronger gravity g** → smaller H : more compressed

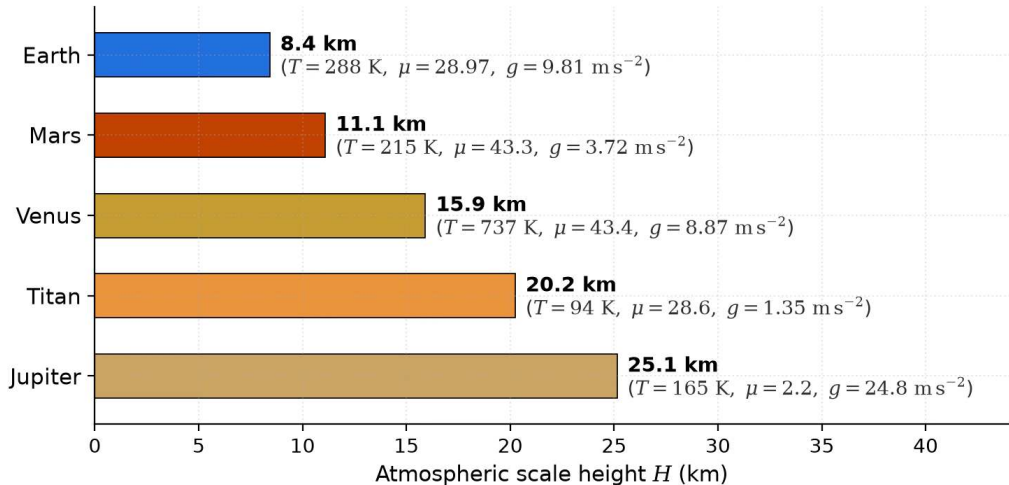
H balances thermal energy against gravitational binding; integrated over the column, the surface pressure is the column weight, $P_0 = m_{\text{col}} g$.

Scale heights across the solar system

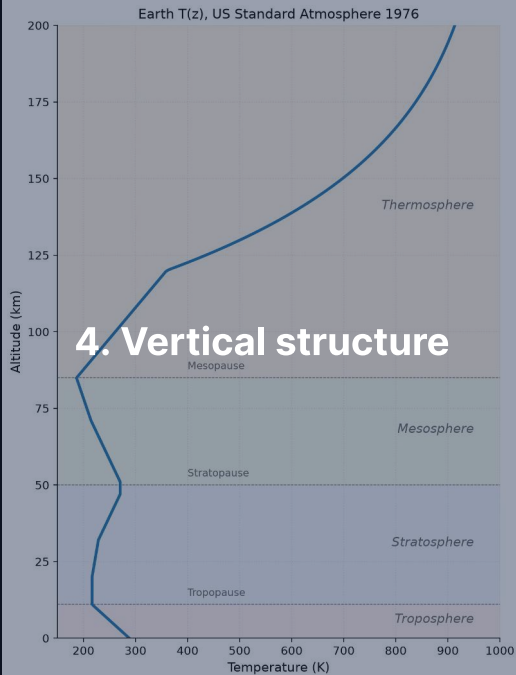
Body	T (K)	μ	g (m s^{-2})	H (km)
Venus	737	43.4	8.87	15.9
Earth	288	28.97	9.81	8.4
Mars	215	43.3	3.72	11.1
Jupiter	165	2.2	24.8	25
Titan	94	28.6	1.35	20

- Jupiter: large H despite strong gravity, because H_2 is very light ($\mu = 2.2$)
- Titan: large H despite low T , thanks to very weak gravity ($g = 1.35 \text{ m s}^{-2}$)
- Venus: hot, but heavy molecules and decent gravity yield a moderate H

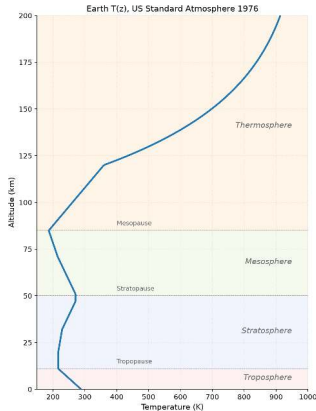
$$\text{Atmospheric scale heights: } H = k_B T / (\mu m_u g)$$



Scale heights recomputed from T , μ and g : light gas (Jupiter) and weak gravity (Titan) give the largest H , Earth the smallest. Course figure.



Earth's atmospheric layers. Course figure



Earth's atmospheric layers, defined by the sign of $\frac{dT}{dz}$. Course figure; US Standard Atmosphere 1976.

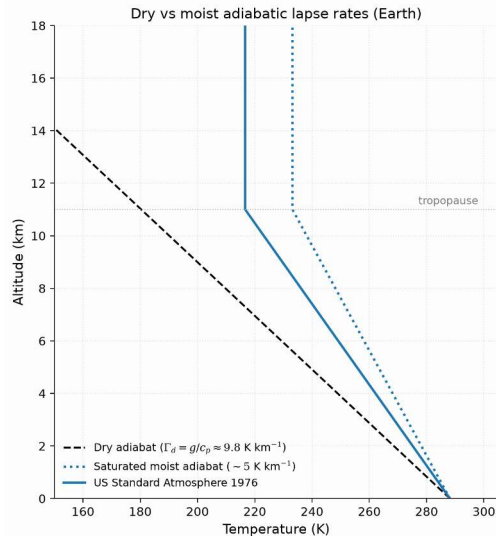
Troposphere (0–12 km): T falls, convective; **stratosphere** (12–50 km): T rises by UV ozone heating; **mesosphere** (50–85 km): T falls to the ~190 K mesopause; **thermosphere** (>85 km): EUV heating, $T > 1000$ K.

The adiabatic lapse rate

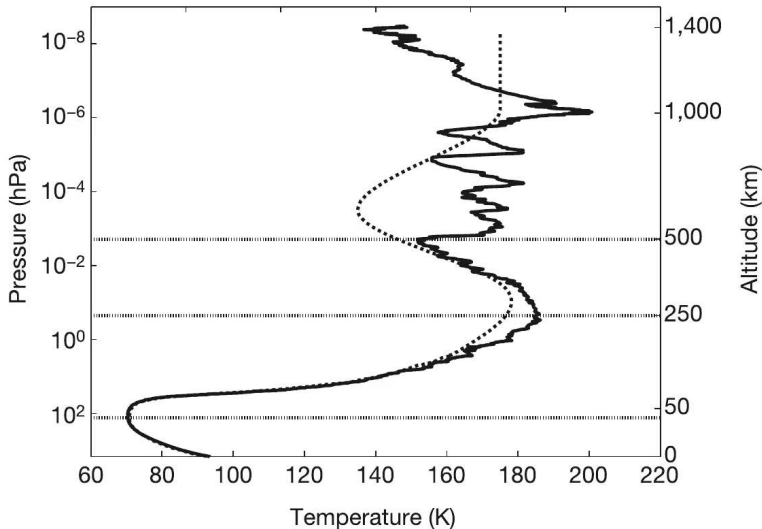
In the troposphere, convection sets the temperature gradient:

$$\Gamma_d = -\frac{dT}{dz} = \frac{g}{c_p}$$

- **Dry adiabatic lapse rate** for Earth: $\Gamma_d = 9.81/1004 \approx 9.8 \text{ K km}^{-1}$; observed $\sim 6.5 \text{ K km}^{-1}$ because of latent heat
- Rising air expands, does work on its surroundings, and cools
- **Tropopause height:** $\sim 8 \text{ km}$ (poles) to $\sim 17 \text{ km}$ (equator)



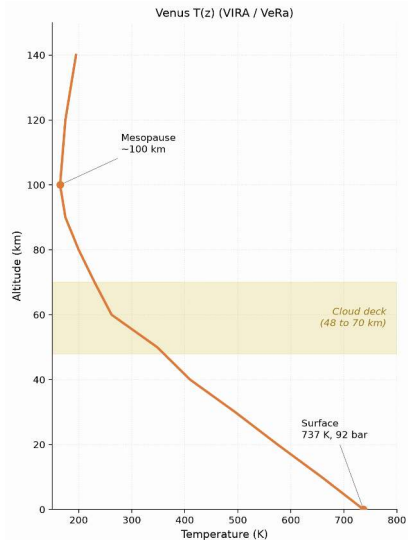
Dry and moist adiabats against the US Standard Atmosphere 1976: latent heat makes the moist adiabat shallower, and real convection sits between both limits. Course figure.



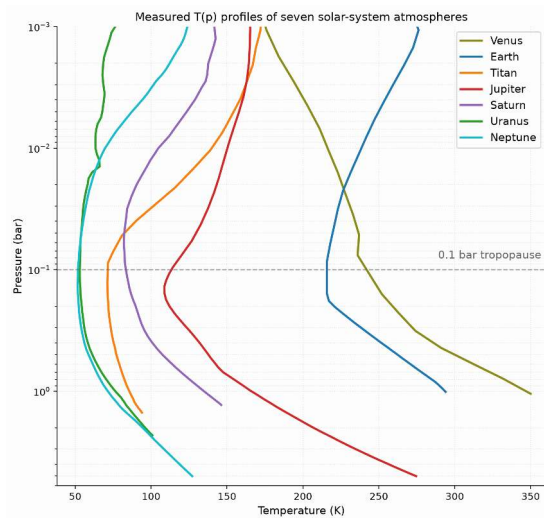
Titan's temperature profile from the Huygens HASI descent: the 94 K surface, the 0.1 bar tropopause and the stratosphere measured in situ. Fulchignoni et al. (2005).



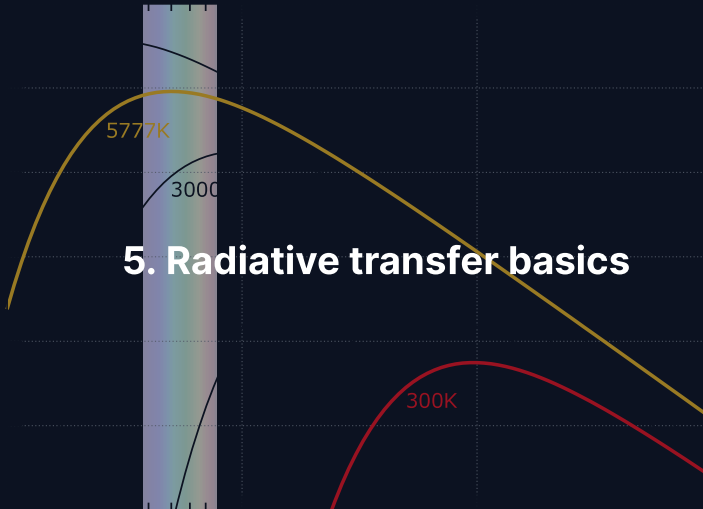
Detached haze layers in Titan's upper atmosphere, Cassini ISS: photochemical haze up to 500 km altitude absorbs solar UV and heats the stratosphere. Credit: NASA/JPL/Space Science Institute, public domain.



Venus temperature profile from Pioneer Venus/VIRA and Venus Express: without ozone, T falls monotonically from 737 K at the surface, reaching 1 bar near 50 km. Course figure.

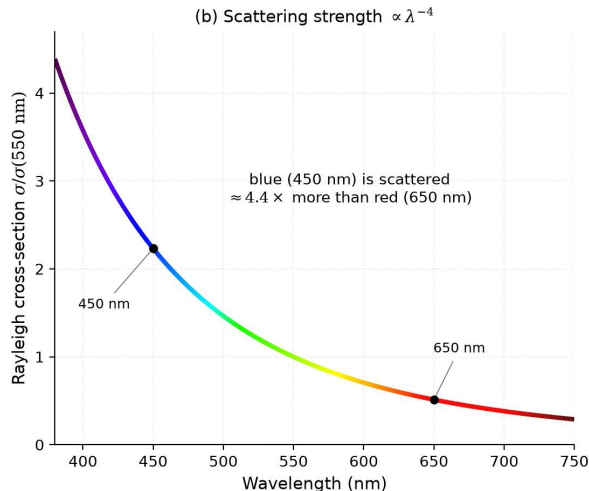
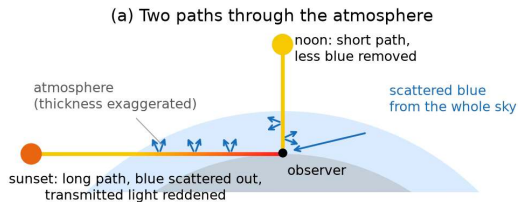


Measured temperature-pressure profiles of seven solar-system atmospheres: thick atmospheres turn from convective to radiative near 0.1 bar, a universal tropopause. Digitized from Robinson & Catling (2014), Fig. 1.

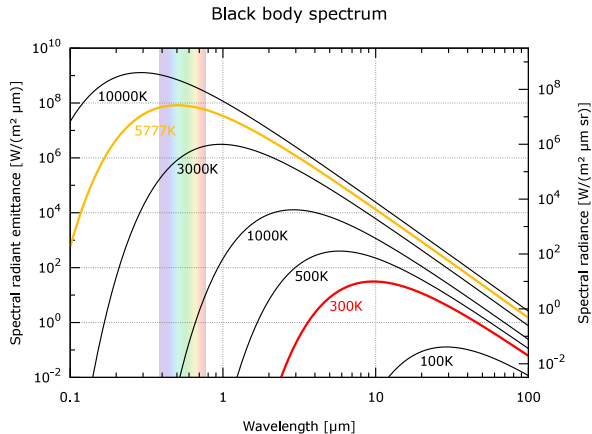


Three interactions: radiation and matter

- **Absorption:** photon energy becomes internal molecular energy; key absorbers CO_2 , H_2O , CH_4 , O_3 , N_2O
- **Emission:** warm gas radiates at the absorbed wavelengths (Kirchhoff's law), directly to space
- **Scattering:** photons redirected without absorption; Rayleigh ($\propto \lambda^{-4}$) and Mie scattering



Rayleigh scattering and the colour of the sky: short paths at noon keep the blue scattered light, long slant paths at sunset transmit only red; the cross-section scales as λ^{-4} . Course figure.



Blackbody spectra of the Sun ($\sim 5800 \text{ K}$, peak $\sim 0.5 \mu\text{m}$) and a planet ($\sim 300 \text{ K}$, peak $\sim 10 \mu\text{m}$).

Credit: Wikimedia, public domain.

This **wavelength separation** is the basis of the greenhouse effect: an atmosphere can be transparent in the visible but opaque in the infrared.

Optical depth and the Beer-Lambert law

Optical depth τ measures the cumulative absorption along a path:

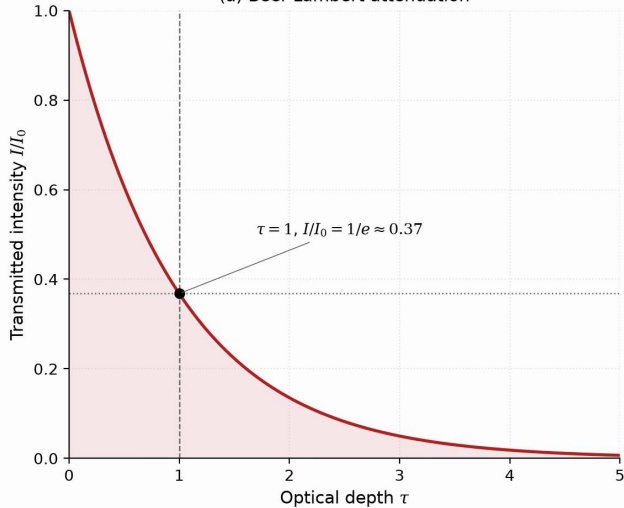
$$\tau = \int_0^s \kappa \rho \, ds'$$

Beam intensity decays following the **Beer-Lambert law**:

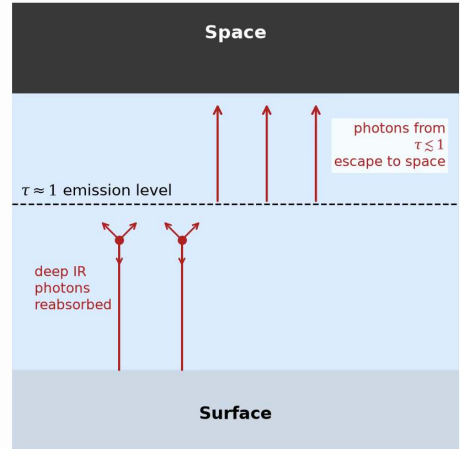
$$I = I_0 e^{-\tau}$$

- **Optically thin** ($\tau \ll 1$): radiation passes freely through the gas
- **Optically thick** ($\tau \gg 1$): photons are absorbed and re-emitted many times
- **Photosphere** ($\tau \approx 1$): thermal radiation escapes directly to space

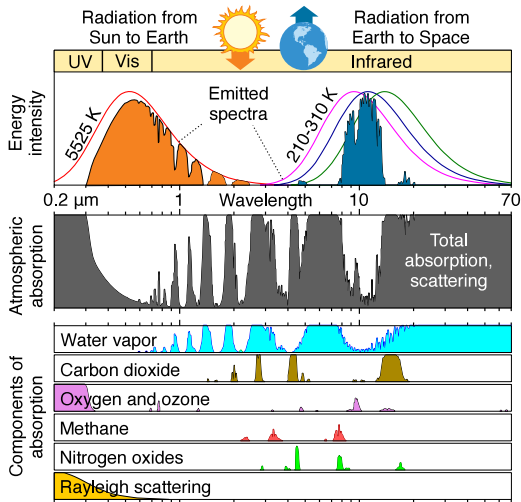
(a) Beer-Lambert attenuation



(b) The atmospheric photosphere

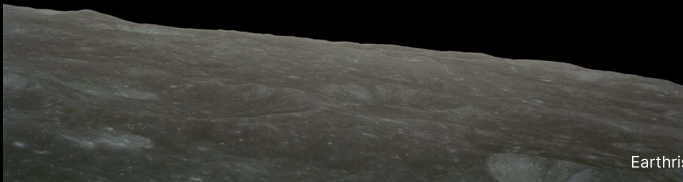


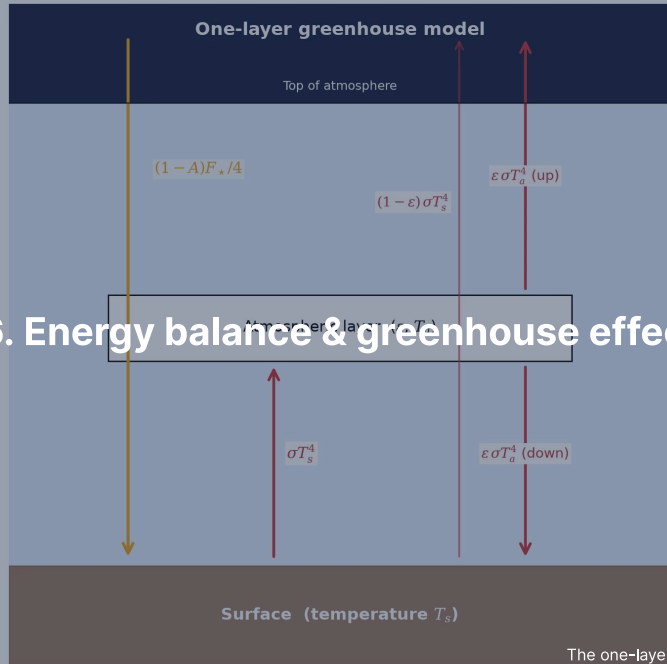
The atmospheric photosphere: thermal radiation escapes to space from the $\tau \approx 1$ level, which sets the effective temperature. Course figure.



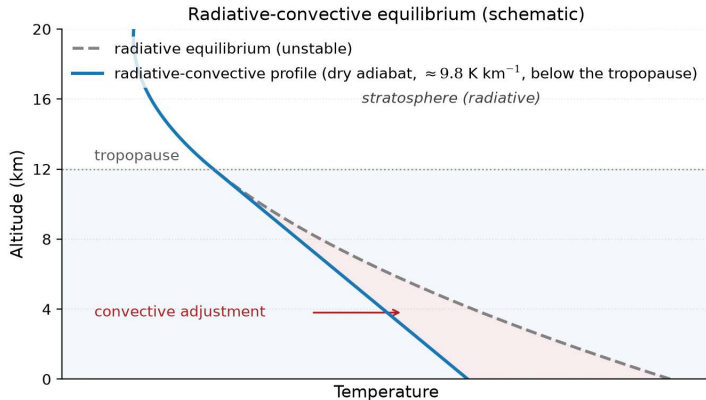
Earth's atmospheric absorption spectrum from UV to IR: greenhouse gases absorb the outgoing infrared while visible sunlight and the 8–13 μm window pass. Credit: Wikimedia Commons, public domain.

Break





6. Energy balance & greenhouse effect



Radiative-convective equilibrium (schematic): the radiative profile is superadiabatic near the surface, convection holds the troposphere on the dry adiabat. Course figure.

Greenhouse opacity makes the **troposphere** convective, the optically thin **stratosphere** sits in radiative equilibrium, and the **tropopause** is where the radiative gradient becomes shallower than the adiabat.

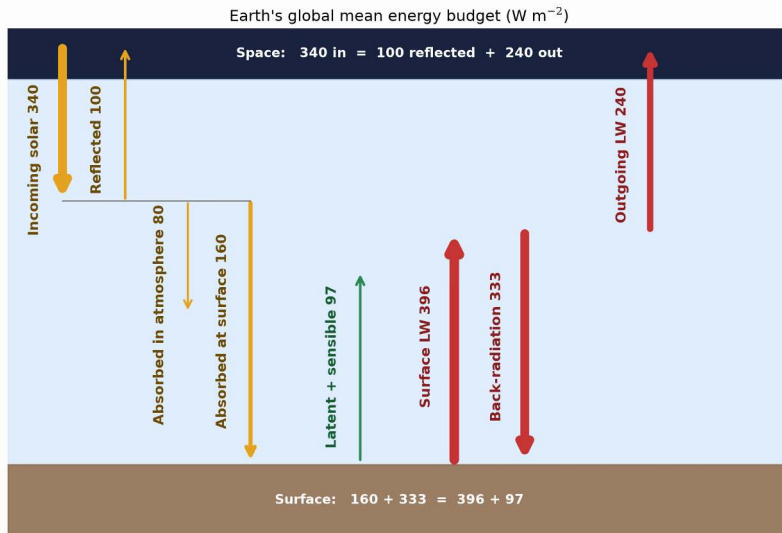
Planetary energy balance

Absorbed stellar power equals emitted thermal radiation:

$$(1 - A) \frac{S}{4} = \sigma T_{\text{eff}}^4 \quad \text{where} \quad S = \frac{L_{\star}}{4\pi d^2}$$

- A: Bond albedo; S: stellar flux ($S_0 = 1361 \text{ W m}^{-2}$ at Earth)
- Factor 1/4: intercepting cross-section πR_p^2 over the emitting sphere $4\pi R_p^2$

$$T_{\text{eff}} = \left[\frac{(1 - A) L_{\star}}{16\pi\sigma d^2} \right]^{1/4} = \left[\frac{(1 - A) S}{4\sigma} \right]^{1/4}$$



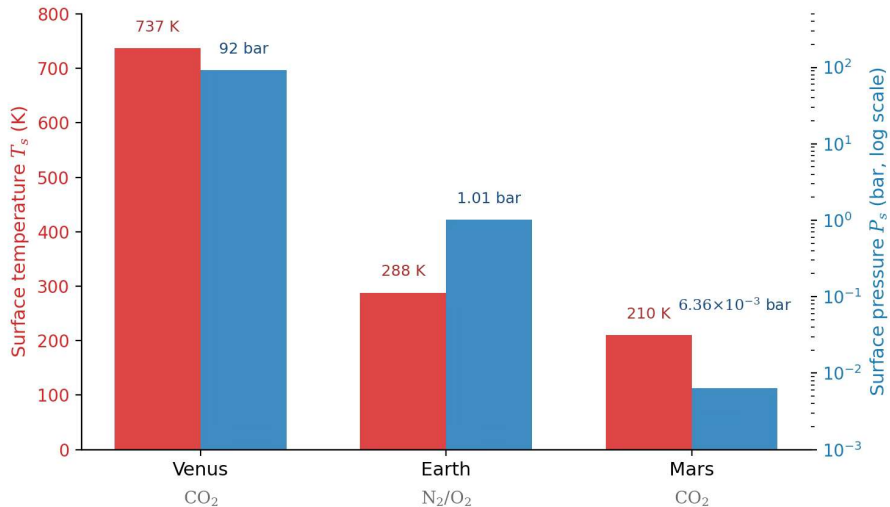
Earth's globally averaged energy budget in W m^{-2} : back-radiation of 333 W m^{-2} supplies more energy to the surface than direct sunlight (160 W m^{-2}). After Trenberth et al. (2009).

Effective vs. surface temperature

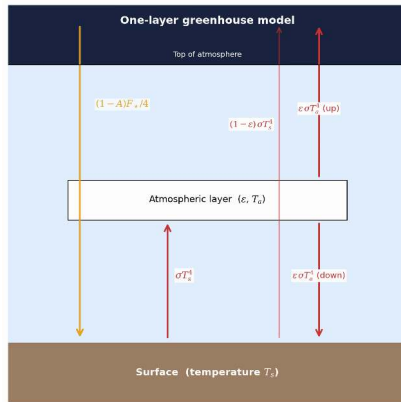
Body	A	d (AU)	T_{eff} (K)	T_{surf} (K)	ΔT (K)
Venus	0.77	0.72	227	737	+510
Earth	0.30	1.00	255	288	+33
Mars	0.25	1.52	210	215	+5
Jupiter	0.34	5.20	110	165*	+55

*Jupiter 1-bar level; excess partly from internal heat (Lecture 3).

$\Delta T = T_{\text{surf}} - T_{\text{eff}}$ is the **greenhouse effect**: 510 K on Venus, 33 K on Earth, ~5 K on Mars (6 mbar, no water-vapour amplifier).



Surface temperature and pressure of Venus, Earth and Mars: similar bulk compositions span a factor 10^4 in pressure and 500 K in temperature through greenhouse warming. Course figure; data from NASA.



The one-layer greenhouse model: sunlight in, infrared absorbed and re-emitted half up, half down. Course figure.

Sunlight ($\sim 0.5 \mu\text{m}$) reaches the surface, the surface emits thermal IR ($\sim 10\text{--}15 \mu\text{m}$), greenhouse gases absorb it and re-emit half downward, and that extra energy makes $T_{\text{surf}} > T_{\text{eff}}$.

One-layer greenhouse model

A single layer with infrared emissivity ε , transparent to visible sunlight:

Flux balance:

$$\text{Atmosphere: } \varepsilon \sigma T_s^4 = 2\varepsilon \sigma T_a^4$$

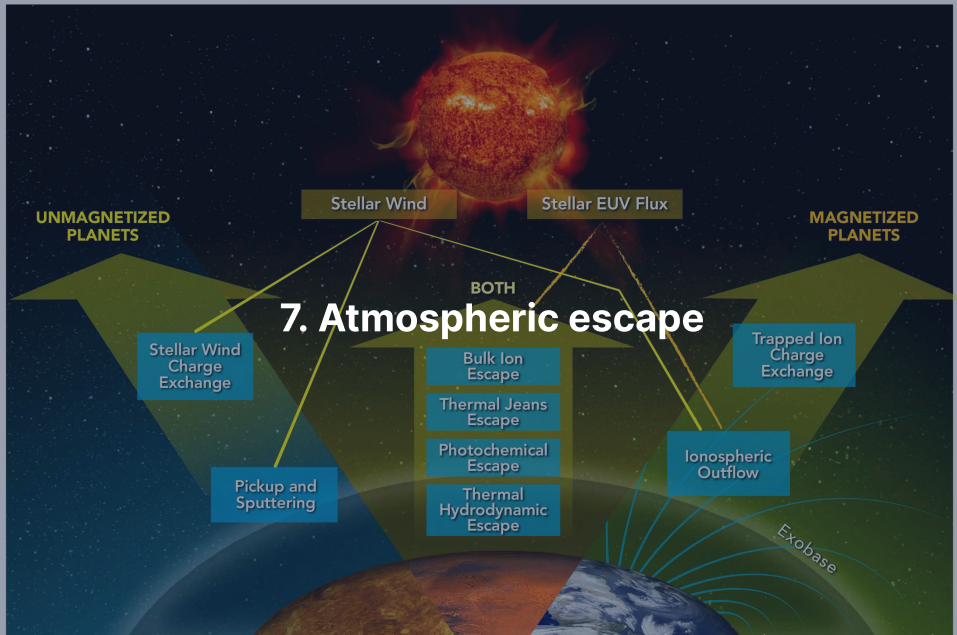
$$\text{Surface: } (1 - A) \frac{F_\star}{4} + \varepsilon \sigma T_a^4 = \sigma T_s^4$$

$$\text{TOA: } (1 - A) \frac{F_\star}{4} = \sigma T_{\text{eff}}^4$$

- $\varepsilon = 0$ (transparent): $T_s = T_{\text{eff}}$
- $\varepsilon = 1$ (opaque): $T_s = 2^{1/4} T_{\text{eff}} \approx 1.19 T_{\text{eff}}$ (Earth: ≈ 303 K)

Surface temperature:

$$T_s = T_{\text{eff}} \left(\frac{2}{2 - \varepsilon} \right)^{1/4}$$

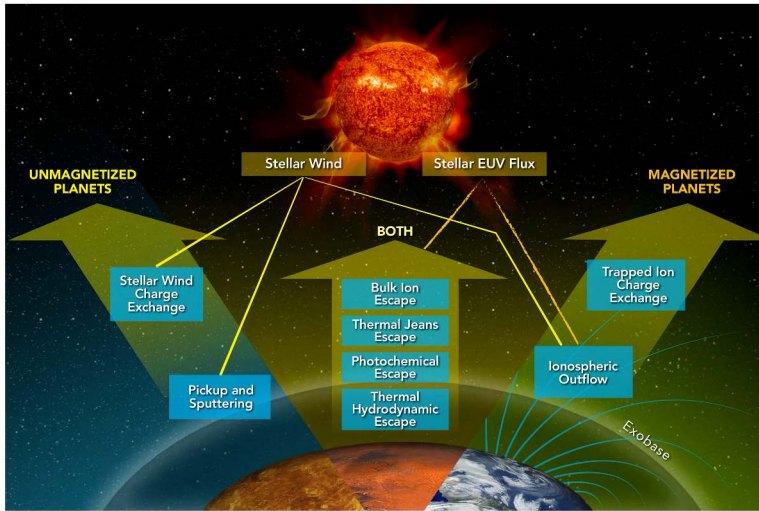


Thermal (Jeans) escape

The high-velocity tail of the Maxwell-Boltzmann distribution can exceed the escape speed. The **Jeans escape parameter** at the exobase compares gravitational to thermal energy:

$$\lambda_J = \frac{GM_p m}{k_B T_{\text{exo}} r_{\text{exo}}} = \frac{v_{\text{esc}}^2}{v_{\text{th}}^2}, \quad v_{\text{th}} = \sqrt{2k_B T_{\text{exo}} / m}$$

- $\lambda_J \gg 1$: escape is negligible because the tail is depleted
- $\lambda_J \lesssim 2-3$: thermal escape drives rapid atmospheric loss
- Escape flux $\Phi_J \propto (1 + \lambda_J) e^{-\lambda_J}$: exponential in λ_J



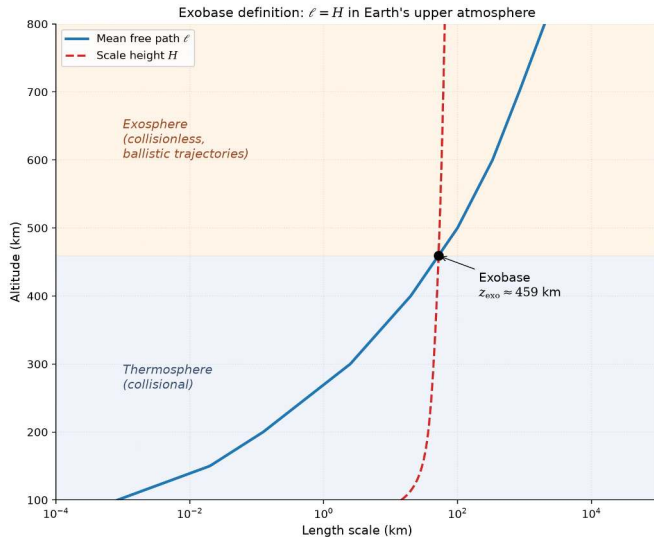
The main atmospheric escape processes: thermal, photochemical and ion channels act on all planets, and the magnetic field selects which dominate. Gronoff et al. (2020).

Jeans parameter for Earth and Mars

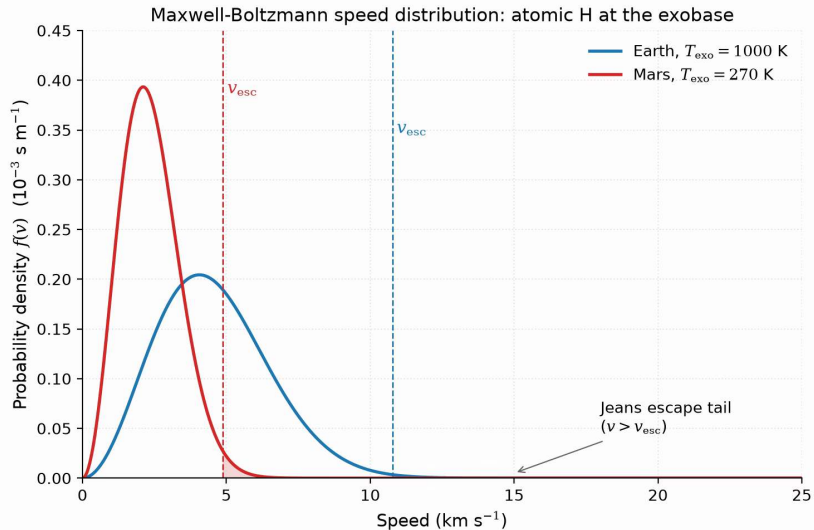
Exobase temperatures: 1000 K (Earth), 270 K (Mars).

Species	m (u)	λ_j (Earth)	λ_j (Mars)
H	1	7.0	5.3
H ₂	2	14	11
He	4	28	21
N ₂	28	200	150
CO ₂	44	310	230

$\lambda_j \propto m$: atomic H escapes steadily from both planets, while N₂ and CO₂ stay gravitationally bound.



The exobase: the altitude where the mean free path ℓ equals the scale height H , collisional fluid below and ballistic flight above. Course figure.



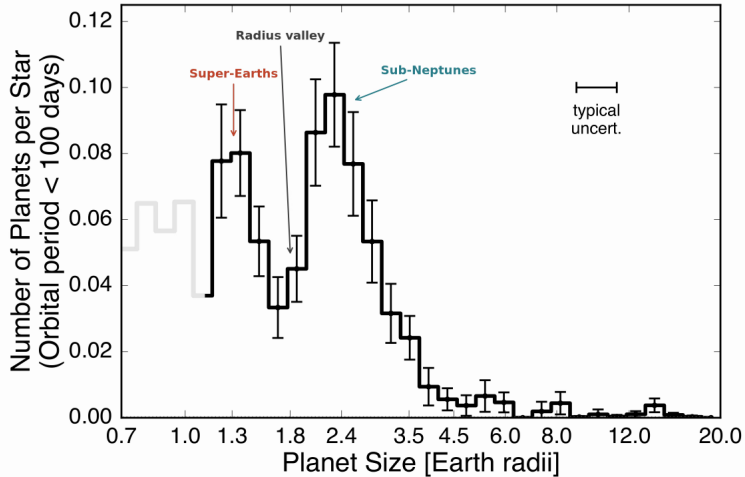
Maxwell-Boltzmann speed distributions for atomic H at the Earth and Mars exobases: the escaping fraction is the tail beyond v_{esc} , scaling as $e^{-\lambda}$. Course figure.

Hydrodynamic escape

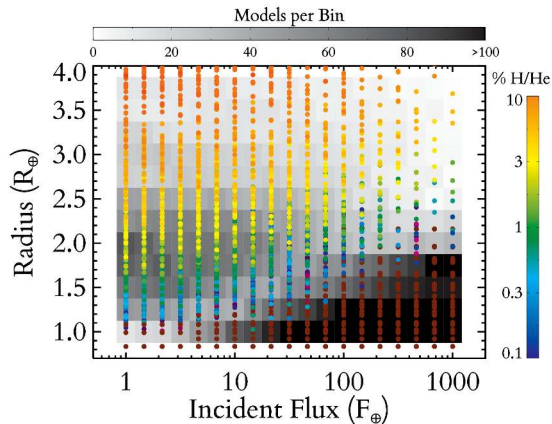
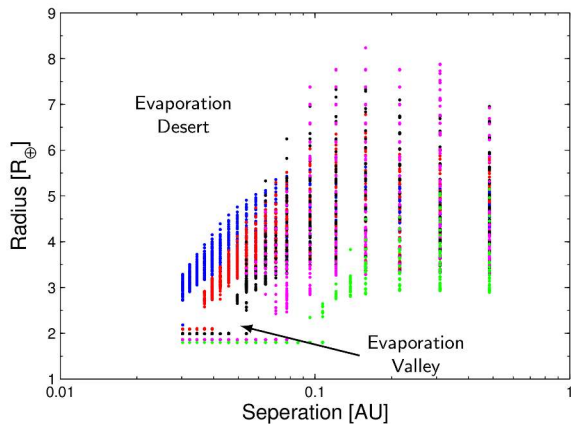
Intense stellar EUV heating drives bulk outflow as an **atmospheric wind**:

- Strongest in the first few hundred Myr, when stellar EUV is 10–100× higher
- Strips H_2 -rich primary envelopes from planets up to several $M_{\oplus}$; outflowing H drags heavier volatiles along
- The primary mechanism behind the exoplanet radius valley at $\sim 1.5\text{--}2 R_{\oplus}$

Source: Hunten et al. (1987); Owen (2019)



The radius valley in the California-Kepler Survey: a deficit near $1.8 R_{\oplus}$ separates stripped rocky cores from envelope-retaining sub-Neptunes. After Fulton et al. (2017).



Population synthesis after 10 Gyr of EUV-driven escape: an empty evaporation desert and a valley near 1.5 to 2 $R_{\oplus}$. After Owen & Wu (2013), Lopez & Fortney (2013) and Owen (2019).

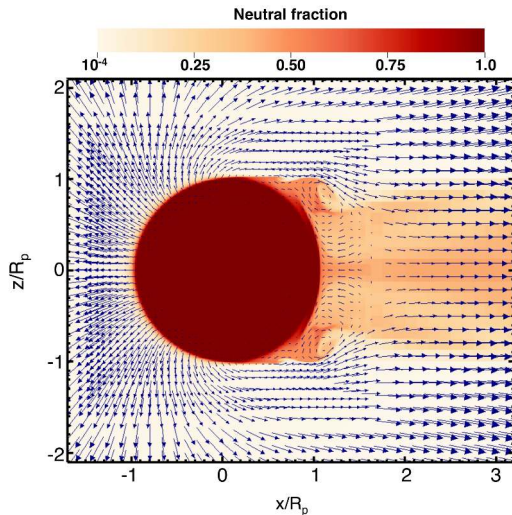
Energy-limited escape rate

Stellar EUV absorbed in the upper atmosphere drives hydrodynamic mass loss:

$$\dot{M}_{\text{EL}} = \eta \frac{\pi F_{\text{EUV}} R_p^3}{G M_p}$$

- Efficiency $\eta \approx 0.1\text{--}0.2$ converts incident EUV flux into expansion work
- Scales as R_p^3 : extended, low-density atmospheres lose mass fastest
- Scales inversely with M_p : low gravity enables stronger bulk outflow

Source: Watson et al. (1981); Owen (2019)



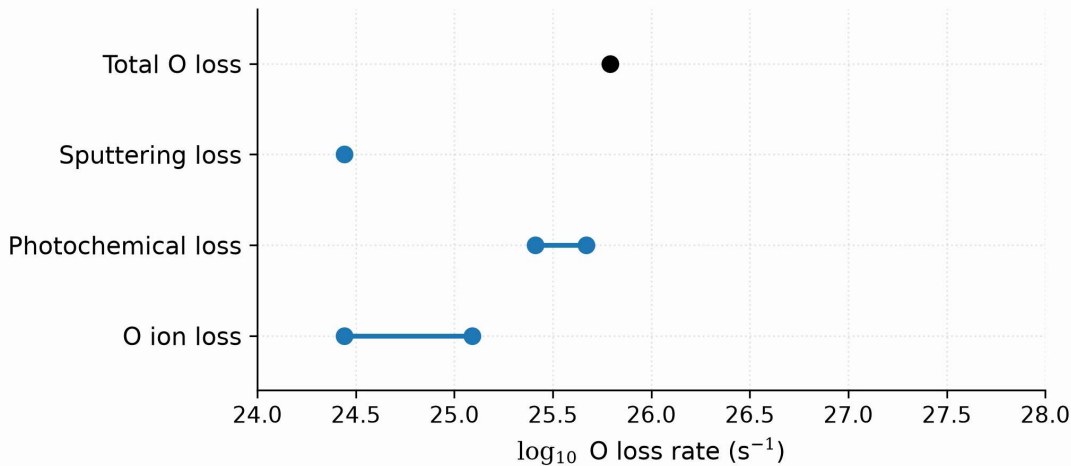
Hydrodynamic outflow from a hot Jupiter under stellar EUV heating: the gas accelerates outward into a night-side wake with Kelvin-Helmholtz instabilities. Reproduced from Owen (2019), Fig. 2; simulation from Tripathi et al. (2015).

Non-thermal escape mechanisms

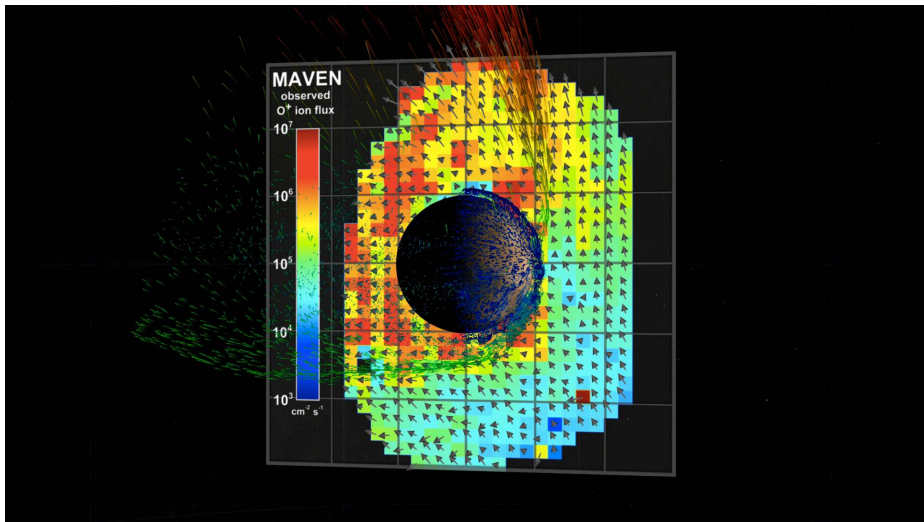
- **Sputtering** and **ion pickup**: solar wind ions eject neutrals, and photoions are swept away by the interplanetary field
- **Photochemical**: photodissociation yields suprathermal atoms above v_{esc}
- **Impact erosion**: giant impacts shock and eject atmospheric columns

MAVEN at Mars: photochemical loss of oxygen dominates modern escape over sputtering and ion pickup.

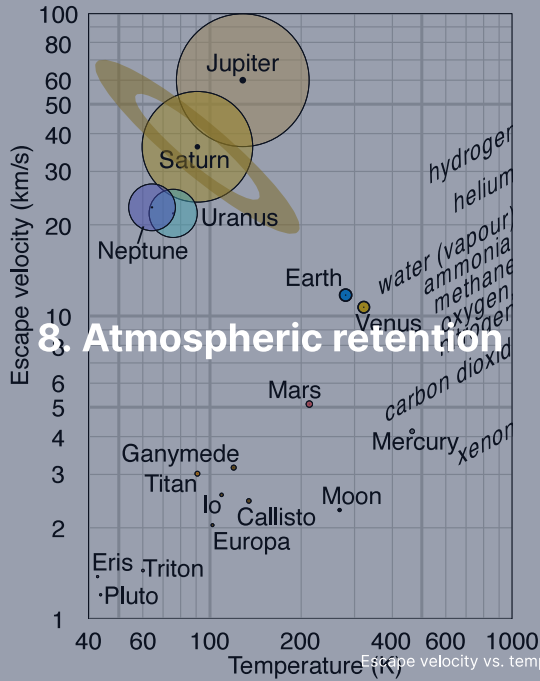
Present-day oxygen loss from Mars

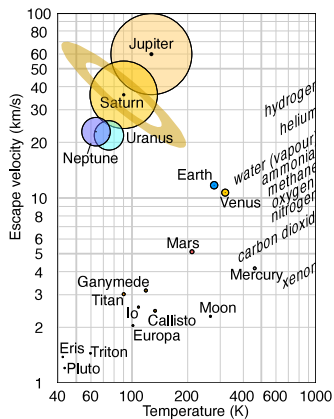


Present-day oxygen loss from Mars by channel: photochemical escape dominates, and with hydrogen escape the total reaches the $2\text{--}3 \text{ kg s}^{-1}$ MAVEN measures. After Jakosky et al. (2018).



The ion plume of Mars from MAVEN: energetic ions (red) leave in a plume, the bulk (green) along the tail, with no global field to keep the solar wind out. Credit: NASA Scientific Visualization Studio (SVS 4370), public domain.





Escape velocity against temperature, with the retention lines for each gas. Credit: Wikimedia, CC BY-SA 4.0.

Long-term retention needs $v_{\text{esc}} \geq 6 v_{\text{th}}$: gas giants keep every species, Earth and Venus lose only H, Mars is marginal, cold Titan keeps N_2 , the Moon and Mercury keep nothing.

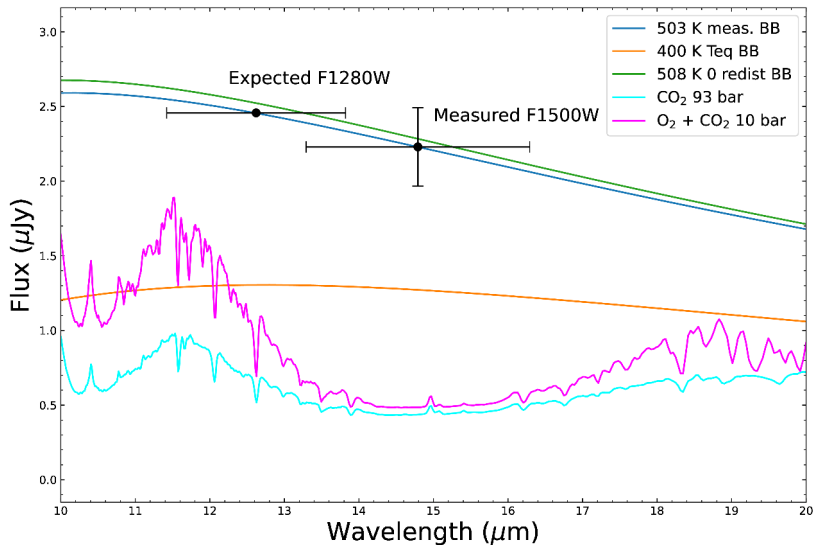


The thin atmosphere of Mars above its limb, Viking 1 (1976). Credit: NASA/JPL.

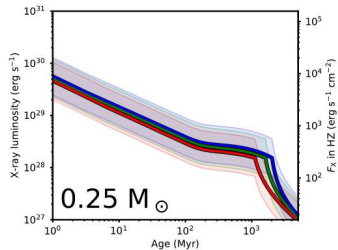
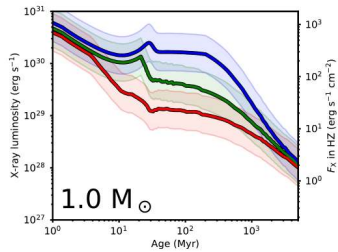
After the dynamo died near ~ 4.1 Ga the solar wind eroded a dense early atmosphere, recorded by valley networks and lakes, down to today's 6 mbar: no stable liquid water, little greenhouse warming.



9. Recent advances & outlook



JWST/MIRI 15 μm thermal emission matches a 503 K bare rock, ruling out a thick atmosphere on TRAPPIST-1 b. Greene et al. (2023).



X-ray luminosity evolution of Sun-like stars and M dwarfs: low-mass stars stay saturated for over 1 Gyr and strip close-in rocky planets for longer. Reproduced from Johnstone et al. (2021), Fig. 11.

Summary

1. **Atmosphere types:** primary (captured), secondary (outgassed), tertiary (biogenic); five orders of magnitude in surface pressure
2. **Hydrostatic balance** and the **scale height** $H = k_B T / (\mu m_u g)$ give $P = P_0 e^{-z/H}$; convection sets $\Gamma_d = g/c_p$ below the 0.1 bar tropopause
3. **Greenhouse warming** from $\tau \gtrsim 1$ raises T_s above T_{eff} ; **atmospheric escape** (thermal, hydrodynamic, non-thermal) is beaten only when $v_{\text{esc}} \gtrsim 6v_{\text{th}}$

Next: Lecture 6

Questions?

Next: Lecture 6, Atmospheres II: clouds, weather & climate (Mara Attia)

Thu 24 Sep, 11:00–13:00, room 5419.0161 (Collegezaal)

Next tutorial: Fri 25 Sep, 13:00–15:00, room 5419.0161 (Collegezaal)

<https://ips.formingworlds.space>